SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 35%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

*I **B. Priyanka (192421206)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
B. Priyanka

Annexure A

Scenario Based Assignment

1.Scenario: A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety .

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

SCENARIO-BASED ASSIGNMENT – ARTIFICIAL INTELLIGENCE

Scenario 1: AI-Powered Drone for Emergency Medicine Delivery

Scenario Description :

A government deploys AI-powered drones to deliver emergency medicines to people living in a flood-affected region. The environment is highly dynamic because roads may suddenly become inaccessible, weather conditions may change unexpectedly, and different terrains may affect travel time and energy consumption.

The drone has access to a heuristic estimate based on straight-line distance, partial real-time updates about blocked paths, and variable movement costs depending on terrain and weather. Due to the urgency of the situation, the drone must quickly identify a route that minimizes delay while maintaining safety.

The main objective is to deliver emergency medicines as quickly and safely as possible. Greedy Best-First Search and A* Search are two informed search algorithms that can help the drone select a suitable route.

1. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

Definition of Greedy Best-First Search

Greedy Best-First Search is an informed search algorithm that selects the next node based on its heuristic value. A heuristic is an estimate of how close a particular location is to the destination.

The algorithm uses the following evaluation function:

\[f(n)=h(n) \]

Where:

- \(f(n)\) represents the evaluation value of node \(n\).
- \(h(n)\) represents the estimated distance or cost from node \(n\) to the destination.

The algorithm always prioritizes the node with the lowest heuristic value. It attempts to move toward the destination as quickly as possible based on the available estimate.

Unlike A*, Greedy Best-First Search does not consider the actual cost already incurred while reaching the current location.

Working of Greedy Best-First Search in the Drone Scenario

Step 1: Identify the starting point and destination

The drone determines its current position and the location of the village where the emergency medicines must be delivered.

Step 2: Calculate heuristic values

The drone calculates the straight-line distance from each possible next location to the destination.

Step 3: Select the most promising location

The algorithm selects the location with the smallest heuristic value, assuming it is the most promising direction.

Step 4: Continue the search

The drone repeats this process at each location, choosing the next location that appears closest to the destination.

Step 5: Respond to blocked paths

If the drone receives information that a route is blocked, it must update its search and consider alternative locations.

Step 6: Reach the destination

The process continues until the drone reaches the destination or determines that it must revise its route.

Example

Suppose a drone must deliver emergency medicines to a village located across a flood-affected region. It identifies two possible routes.

- Route A: A shorter route that appears close to the destination but passes through an area affected by strong winds.
- Route B: A longer route with more favourable weather conditions.

Greedy Best-First Search may select Route A because its estimated remaining distance is smaller. However, strong winds may increase the actual travel time or make the route unsafe.

If Route A becomes inaccessible, the drone must update its route information and search for another path.

This example shows that a route that appears closest to the destination is not necessarily the fastest or safest route.

Strengths of Greedy Best-First Search

1. Fast decision-making

Greedy Best-First Search prioritizes locations that appear closer to the destination. This can help the drone identify a promising direction quickly, which is important during emergency medicine delivery.

2. Simple implementation

The algorithm uses a heuristic estimate to prioritize locations. This makes it relatively easy to implement and understand.

3. Reduced exploration in suitable environments

When the heuristic provides useful guidance, the algorithm may explore fewer locations than uninformed search algorithms.

4. Useful under time constraints

The algorithm can quickly identify a promising route when the drone must make decisions within a short period.

5. Can respond to updated information

If the search is updated after the drone discovers a blocked route, the algorithm can explore alternative locations.

Weaknesses of Greedy Best-First Search

1. Ignores the actual path cost

The algorithm does not consider the accumulated cost of reaching the current location. A route may appear short but require more time or energy.

2. Does not guarantee an optimal solution

The algorithm may find a route quickly, but it does not guarantee that the route has the lowest total travel cost.

3. Sensitive to inaccurate heuristics

If straight-line distance does not reflect actual travel conditions, the drone may choose an unsuitable route.

4. Difficulties under uncertainty

The drone may not have complete information about weather, obstacles, or inaccessible routes. Its initial decision may therefore be based on incomplete information.

5. Safety concerns

A short route may pass through hazardous terrain or dangerous weather. The algorithm does not automatically prioritize safety unless additional safety restrictions are introduced.

Conclusion

Greedy Best-First Search is useful when a quick, approximate route is required. However, because it ignores accumulated travel cost and does not guarantee an optimal route, it is less suitable as the sole planning algorithm for safety-critical emergency medicine delivery.

2. Explain how A* would handle the same situation, including how it balances cost and heuristic.

Definition of A* Search

A* is an informed search algorithm that considers both the actual cost of reaching a location and the estimated cost of travelling from that location to the destination.

It uses the following evaluation function:

$$ f(n)=g(n)+h(n) $$

Where:

- $f(n)$ represents the estimated total cost of the route through node n.
- $g(n)$ represents the actual accumulated cost from the starting point to node n.
- $h(n)$ represents the estimated remaining cost from node n to the destination.

Unlike Greedy Best-First Search, which considers only the estimated remaining cost, A* considers both the accumulated cost and the estimated remaining cost.

Working of A* Search in the Drone Scenario

Step 1: Identify the starting point and destination

The drone identifies its current position, the destination, and the available routes connecting them.

Step 2: Calculate the accumulated cost

The drone calculates the cost of reaching each possible location. This may include travel time, energy consumption, terrain difficulty, and penalties for hazardous conditions.

Step 3: Estimate the remaining cost

The drone uses a heuristic to estimate the cost of travelling from each candidate location to the destination.

Step 4: Calculate the evaluation function

For each candidate location, the drone calculates:

$$ f(n)=g(n)+h(n) $$

The algorithm prioritizes the candidate with the lowest estimated total cost.

Step 5: Continue searching

The drone explores further locations and updates the costs of possible routes as it discovers more information.

Step 6: Update the route when necessary

If a route becomes blocked, the drone updates its map, removes or invalidates the affected path, and searches for another feasible route.

Step 7: Reach the destination

The process continues until the drone identifies a suitable route to the destination.

Example of A* Cost Calculation

Suppose the drone has two possible routes to deliver medicines.

Factor	**Route A**	**Route B**
Actual cost so far, $g(n)$	8	3
Estimated remaining cost, $h(n)$	2	6
Total estimated cost, $f(n)$	10	9

The values represent illustrative cost units.

For Route A:

$$ f(A)=8+2=10 $$

For Route B:

$$ f(B)=3+6=9 $$

Although Route A appears closer to the destination, Route B has the lower estimated total cost. A* therefore prioritizes Route B.

This example demonstrates how A* balances the cost already incurred with the estimated cost remaining.

Strengths of A* Search

1. Considers actual and estimated costs

A* considers both the cost of reaching the current location and the estimated cost of completing the route. This makes it more suitable for situations involving variable travel costs.

2. Can find an optimal route

A* can find a minimum-cost route when the cost model is appropriate, the search conditions are satisfied, and the heuristic is admissible. An admissible heuristic never overestimates the true remaining cost.

3. Can account for weather and terrain

The cost function can include the effects of wind, terrain difficulty, energy consumption, and other relevant factors.

4. Reduces unnecessary exploration

A useful heuristic directs the algorithm toward promising routes instead of exploring every possible location equally.

5. Supports route replanning

When new information becomes available, the search can be updated to identify another route.

Weaknesses of A* Search

1. High memory requirements

A* may need to store many candidate locations and their associated costs. Large maps can therefore consume significant memory.

2. Computational cost

If the search space is large, evaluating multiple possible routes may require considerable processing time.

3. Dependence on heuristic quality

An inaccurate heuristic may reduce efficiency. If the heuristic overestimates the remaining cost, A* may lose its optimality guarantee.

4. Dynamic environmental changes

Frequent changes in weather and route accessibility may require the drone to update its calculations and replan.

5. Safety is not automatic

A* minimizes the cost represented in its model. If the model does not properly account for hazards, the algorithm may still select an unsafe route.

Conclusion

A* is more suitable than Greedy Best-First Search when the drone must consider travel cost, changing environmental conditions, and route efficiency. However, it must be combined with reliable information, appropriate cost modelling, and explicit safety constraints.

3. Analyze the impact of heuristic accuracy on both algorithms.

Definition of a Heuristic

A heuristic is an estimate of the remaining cost from a current location to the destination. In this scenario, straight-line distance is used to estimate how far the drone is from the village.

Heuristic accuracy affects the quality of the selected route and the amount of computation required.

Impact on Greedy Best-First Search

1. Accurate heuristic

When straight-line distance provides useful guidance, the algorithm can prioritize locations that move the drone toward the destination.

2. Inaccurate heuristic

If the shortest-looking direction contains obstacles or difficult terrain, the algorithm may choose an unsuitable route.

3. Ignores accumulated cost

Even with a good heuristic, Greedy Best-First Search may select a route that has already required considerable travel time or energy.

4. No optimality guarantee

The algorithm does not guarantee a minimum-cost route because it considers only the estimated remaining cost.

Impact on A* Search

1. Accurate heuristic

A useful heuristic guides A* toward promising routes and may reduce unnecessary exploration.

2. Admissible heuristic

If the heuristic never overestimates the true remaining cost, A* can preserve its optimality guarantee under the appropriate search conditions.

3. Overestimated heuristic

If the heuristic overestimates the remaining cost, A* may select a suboptimal route and lose its optimality guarantee.

4. Underestimated heuristic

An underestimated heuristic can preserve optimality when it remains admissible, although a very weak heuristic may cause A* to explore more locations.

5. Weather and terrain effects

Straight-line distance may be appropriate for estimating remaining distance, but it does not necessarily estimate remaining travel time, energy consumption, or risk accurately.

Comparison of Heuristic Accuracy in Greedy Best-First Search and A* Search :

Heuristic accuracy refers to how closely the estimated cost matches the actual cost required to reach the destination. In this scenario, the drone uses straight-line distance to estimate how far it is from the village where emergency medicines must be delivered.

1. Accurate Heuristic

In Greedy Best-First Search, an accurate heuristic helps the drone select locations that appear closer to the destination. However, the algorithm ignores the actual cost of reaching those locations.

In A* Search, an accurate heuristic helps guide the drone toward the destination while considering the actual cost already incurred. This can improve search efficiency and route selection.

2. Inaccurate Heuristic

In Greedy Best-First Search, an inaccurate heuristic may cause the drone to select a route that appears shorter but is blocked or affected by difficult terrain.

In A* Search, an inaccurate heuristic can also affect route selection. However, A* considers the accumulated cost along with the heuristic, which may help it avoid some inefficient routes.

3. Overestimated Heuristic

An overestimated heuristic predicts that the remaining cost is greater than it actually is.

In Greedy Best-First Search, this can mislead the algorithm because it relies entirely on the estimated remaining cost.

In A* Search, an overestimated heuristic may cause the algorithm to lose its guarantee of finding the minimum-cost route.

4. Underestimated Heuristic

An underestimated heuristic predicts a remaining cost that is lower than the actual cost.

Greedy Best-First Search may still choose an inefficient route because it ignores the cost already incurred.

A* Search can preserve its optimality guarantee when the heuristic never overestimates the true remaining cost and the other required conditions are satisfied. However, a very weak heuristic may cause A* to explore more locations and take longer to find a solution.

5. Effect of Weather and Terrain

Straight-line distance does not account for strong winds, dangerous terrain, or blocked paths.

Greedy Best-First Search may select a short but unsuitable route if these factors are not reflected in its heuristic.

A* Search can consider weather conditions, terrain difficulty, and travel time through its cost function. This helps it identify a route with a lower total estimated cost.

Conclusion:

Heuristic accuracy affects both algorithms, but A* Search is generally more suitable for cost-sensitive route planning because it considers both the actual accumulated cost and the estimated remaining cost. For emergency medicine delivery, the heuristic and cost function should reflect travel time and relevant environmental conditions rather than distance alone.

Conclusion

Heuristic accuracy affects both algorithms. However, A* is generally more appropriate for cost-sensitive planning because it considers both accumulated cost and estimated remaining cost. A suitable heuristic should reflect the actual optimization objective, such as travel time, rather than distance alone.

4. Discuss how dynamic changes, such as blocked paths, affect search performance.

The drone operates in a flood-affected environment where routes and weather conditions can change unexpectedly. These changes may make previously calculated routes unsuitable and require the search algorithm to update its decisions.

Major Challenges

1. Blocked routes

A route that was initially available may become inaccessible because of flooding, debris, or other obstacles. The drone must identify another feasible route.

2. Changing weather conditions

Strong winds, rain, or storms may increase travel time, energy consumption, or operational risk.

3. Incomplete real-time information

The drone may not receive immediate information about every obstacle or weather change. It may therefore make decisions using outdated information.

4. Increased computation

When route conditions change, the algorithm may need to update its search results. Frequent changes can increase computation time.

5. Delivery delays

Repeated replanning may delay medicine delivery. This is especially important when patients require urgent medical assistance.

Effect on Greedy Best-First Search

Greedy Best-First Search can quickly prioritize another location when route information is updated. However, it may choose a replacement route that appears close but has a high actual travel cost.

Since it does not consider accumulated cost, it may repeatedly select promising-looking routes that are inefficient in practice.

Effect on A* Search

A* can compare alternative routes using their accumulated and estimated costs. It can account for changing conditions when route costs and available paths are updated.

However, repeated changes may require additional computation, especially when the map is large or the environment changes rapidly.

Improving Performance Through Dynamic Replanning

An incremental search algorithm such as D* Lite can help the drone respond to environmental changes by reusing previous search information instead of planning the entire route from scratch.

The system should:

- Continuously update information about blocked routes and weather.
- Recalculate costs when conditions change.
- Reject routes that violate safety constraints.
- Select an alternative route when the current route becomes infeasible.
- Monitor battery levels and maintain sufficient energy reserves.
- Reassess the route when new hazards are detected.

Conclusion

Dynamic changes reduce the reliability of fixed route plans. A* with suitable replanning mechanisms can adapt to updated information and help the drone maintain an efficient and feasible route.

5. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

Recommended Approach: A* with Dynamic Replanning and Explicit Safety Constraints

A* is the most suitable starting point for this scenario because the drone must consider accumulated travel cost, estimated remaining cost, and changing environmental conditions.

Greedy Best-First Search may find a promising route quickly, but it ignores the actual cost of reaching the current location. A* provides a better balance between route efficiency and estimated remaining cost.

Justification

1. Real-time constraints

The drone must make decisions quickly because emergency medicines need to reach patients without unnecessary delays. A* uses a heuristic to guide the search toward promising routes. Efficient implementation and appropriate search limits can help it produce decisions within the required time.

2. Optimality

A* can find a minimum-cost route when the cost model is suitable and the heuristic satisfies the required conditions. This makes it preferable when the total travel cost matters, rather than distance alone.

3. Safety

The route cost can account for hazards, difficult terrain, weather, and energy consumption. However, safety-critical requirements should be implemented as hard constraints rather than relying only on cost penalties. Routes that violate safety limits must be rejected.

4. Adaptability

When a route becomes blocked, the drone can update its map and replan. D* Lite or another incremental replanning method can reuse previous search information and reduce repeated computation.

5. Efficient medicine delivery

By considering accumulated and estimated costs, A* can identify routes that balance travel time, energy use, and risk more effectively than an algorithm that considers only distance to the destination.

6. Handling uncertainty

The drone may receive incomplete or delayed information about environmental conditions. It should therefore update its internal map as new observations become available and revise its route when necessary.

A* can support this process when combined with an appropriate replanning strategy. However, no search algorithm can guarantee safety if its environmental information is inaccurate or important hazards are not represented in its model.

Final Recommendation

A* combined with dynamic replanning and explicit safety constraints is the recommended approach for the emergency medicine delivery drone.

It offers a suitable balance between route efficiency, optimality, and adaptability. Although A* may require more memory and computation than Greedy Best-First Search, its ability to consider actual and estimated route costs makes it more appropriate for this safety-critical application.

The drone should prioritize routes that are safe and feasible, even if they are not the shortest. If conditions change unexpectedly, it should update its information and replan promptly. This combination allows the drone to respond to uncertainty while minimizing avoidable delays in delivering emergency medicines.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand